

Zimu Wang

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Education

University of California, Berkeley, Berkeley, CA

Graduation: May 2027

- Bachelor of Arts in Computer Science, GPA: 3.92/4.0
- Coursework: Neural Network, Deep Reinforcement Learning, Computer Vision, Optimization, Probability Theory, Databases, Data Structures & Algorithms

Experience

ByteDance, LLM Engineering Intern

Summer 2026

- Primary author of **RSIHub** (github.com/simple-agent-lab/RSIHub), a **recursive self-improvement** framework that evolves agent prompts, skills, harnesses, and code.
- Designed a modular evolution engine isolating mutable agent components from fixed evaluators, with every generation tracked in Git so each result can be reproduced from its commit.
- Built a plugin-style operator SDK with automatic discovery, subprocess isolation, and declarative config schemas, letting new strategies ship without modifying core.
- Benchmarked four self-improvement strategies across MiniSWE and Codex agent harnesses on the 89-task Terminal-Bench 2 and 97-task Tau3-Bench Banking suites; best full-benchmark gains were **+13.5 pts** on Terminal-Bench 2 with MiniSWE (56.2% → 69.7%) and **+9.3 pts** on Tau3 Banking with Codex (24.7% → 34.0%).
- Built a lazy-loading skill system and modular agent starter for **Simple Long Horizon Agent** (github.com/simple-agent-lab/simple-long-horizon-agent), unifying four agent variants under a shared session and tool lifecycle with config-driven MCP support.

Shanghai Jiao Tong University, Researcher

Summer 2025 – Spring 2026

- Developed **EffiSkill**, a two-stage LLM framework that mines reusable optimization skills from slow/optimized program pairs and applies them to unseen programs for execution-free rewriting.
- Built inference and evaluation pipelines on **EffiBench-X** (623 tasks, Python and C++), improving optimization success rate over the strongest baseline by **3.69–12.52 percentage points**.
- Built an end-to-end **SFT + GRPO** post-training pipeline for Qwen3-8B code judges on 48K examples with verl/vLLM on **8×A100**, implementing custom reward functions, distributed rollouts, and held-out evaluation; lifted agreement with ground truth from 54% to **69%**, surpassing **Qwen3-32B (66%)** with one-quarter the parameters.

University of California, Berkeley, Data 8 Tutor

Fall 2024

- Led weekly tutoring sessions, a discussion section, and office hours for Data 8, covering Python, statistical inference, and data analysis.

Publication

- *EffiSkill: Agent Skill Based Automated Code Efficiency Optimization* [arXiv]
Zimu Wang, Yuling Shi, Mengfan Li, Zijun Liu, Jie M. Zhang, Chengcheng Wan, Xiaodong Gu.
Under ACL Rolling Review; targeting EACL 2027.

Skills

Languages: Python, C++, Java, SQL, Bash

ML/Agents: PyTorch, Hugging Face Transformers, verl, vLLM, SFT, GRPO, RLHF, MCP, agent evaluation

Infra/Tools: Docker, Weights & Biases, Git, Linux, NumPy, Pandas